# What Predicts Breaking Through Strength Plateaus?

Have you ever found it hard for yourself to progress in the gym? If so, you may have been experiencing a strength plateau. A phenomenon that only gets more common as you gain more experience weightlifting. Renner et al. (2026) found that powerlifters typically reach their peak strength at age 27 after four to six years of competition, using age, sex, competition frequency, and weight class change as predictors. Ferrari et al. (2025) built a machine learning-based linear regression model to predict overall performance in classic powerlifters using the same variables, but with the addition of initial strength levels and competitive history. Latella et al. (2024) modeled changes in strength over time, examining the rate and magnitude of strength progression based on the same variables. None of these studies examined what happens after a plateau begins. Does the lifter remain at the plateau, or do they break through and hit a new personal best, and what is the most reliable predictor(s) of this breakthrough? Using OpenPowerlifting.org's public data which contains 4,000,000+ entries for 1,000,000+ athletes across years of competition, this project will identify plateau periods, and build a logistic regression model that predicts whether or not an athlete will break out of a plateau, using age, competition frequency, and weight-class changes as predictors. The output of this project will be in the form of a research paper, and code notebook with the accompanying model. The intellectual merit of this project lies in doing research around recovering from a plateau, not doing research on the onset of one as the current research does. The broader impact of this project is practical and can be helpful to competitive athletes and coaches who want to find evidence-based answers for how to break out of a strength plateau.

To complete this project, I will primarily use python to not only clean and prepare my data, but perform the bulk of the analysis. OpenPowerlifting.org provides data in a csv format,

which is easily imported into python. For my statistical analysis, I will likely use the pandas library in python, and some visualization libraries like seaborn. My coursework in applied statistics and econometrics has given me some knowledge on regression, but I will likely have to do more research into logistic regression which I will do through the use of textbooks supplemented with online tutorials. To make sure my model is correct and makes sense, I will go to office hours as the semester progresses.

My experience as a competitive swimmer and as someone who enjoys weight lifting has given me first hand experience into strength plateaus. Latella et al. (2024) and Renner et al. (2026) show that the most strength gains happen in an athlete's early career and diminish over time. Steele et al. (2022) studied long term strength gains over time using minimal-dose resistance training and found that strength progression is not linear, and interestingly enough, that age, sex, and bodyweight had minimal interaction effects. None of these studies tests what distinguishes between lifters who break out of their plateau and lifters who do not. This project takes the large-scale statistical approach that Latella and Renner take and applies it to the recovering from a plateau question that the current research does not cover.

I plan to do my data acquisition and cleaning for this project this coming week. After that, I will hopefully be able to define and validate plateau/breakthrough thresholds that will shape the results of my research. Next, I will perform some exploratory analysis into the data followed by an initial logistic regression model which will hopefully be done by mid October. By early November, I hope to be able to interpret my results, followed by a written report draft in mid November. This should adequately prepare me for my final deliverable due in Early December.

Sources:

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